

Technical Report: PGA Tournament Success Prediction - Driving vs Putting

1. Introduction

The goal of this project is to develop a predictive model that determines factors contributing to success in PGA tournaments. Specifically, the focus is on evaluating whether driving performance or putting performance has a greater impact on golf tournaments.

2. Data Collection and Preprocessing

The dataset consists of 1,387 observations from 19 PGA Tour performance statistics across multiple tournaments during the 2022 season. The dataset is from [SCORE Sports Data Repository](#). The analysis was conducted in Python using pandas, numpy, and scikit-learn. Data was split into training and testing sets using an 80/20 split.

Key variables include:

- Driving Distance
- Driving Accuracy
- Putting Statistics
- Scoring Metrics

Preprocessing Steps

1. Data Cleaning

- Removed missing or inconsistent values.
- Standardized column formats for modeling.

2. Feature Selection

- Selected performance-based metrics relevant to tournament success.
- Removed highly correlated features when needed.

3. Feature Scaling

- Applied normalization/standardization where appropriate.

4. Train-Test Split

- Data was split into training and testing sets to evaluate generalization performance.

3. Modeling Approach

Multiple machine learning models were implemented to compare performance:
Models Used

- Logistic Regression
- Decision Tree
- Gradient Boosting

Each model was chosen for different strengths:

- Logistic Regression: Baseline linear model for interpretability
- Decision Tree: Captures nonlinear relationships
- Gradient Boosting: Strong predictive performance through sequential learning

We used gradient boosting as our new model because it builds models sequentially, where each new decision tree focuses on correcting the errors of the previous ones. By using a gradient descent, this minimizes prediction error and allows the model to capture the complex patterns in the dataset.

4. Experimentation

Each model was tested with different parameter configurations:

- Decision Tree: depth, splitting criteria
- Gradient Boosting: learning rate, number of estimators

This experimentation ensured deeper analysis beyond default settings.

Additional features were explored, including:

- Aggregated scoring metrics.
- Composite performance indicators.

These features aimed to better capture overall player performance.
A key extension beyond class material was the use of:

- Gradient Boosting, which improves performance through iterative error correction
- ROC-AUC evaluation, providing a more robust metric than simple accuracy

5. Evaluation Metrics

Models were evaluated using:

- Accuracy: Overall prediction correctness
- ROC-AUC Score: Ability to distinguish between successful vs. unsuccessful outcomes
- Cross-Validation Mean Scores: Stability across multiple data splits

Why These Metrics?

- Accuracy alone can be misleading if classes are imbalanced
- ROC-AUC provides a better measure of model discrimination
- Cross-validation ensures robustness and avoids overfitting

6. Results and Comparison

Across all models:

- Gradient Boosting achieved the highest predictive performance (Accuracy: 0.788, ROC-AUC: 0.746)
- Decision Tree showed more variance and overfitting (ROC-AUC: 0.642)
- Logistic Regression provided strong performance (ROC-AUC: 0.756) while maintaining interpretability.

Key Findings

Although driving metrics slightly improve predictive accuracy, putting metrics consistently showed stronger importance than driving metrics across models and have a greater impact on performance.

This highlights that putting stats such as putts per round are a bigger predictor of success than driving metrics such as driving distance.

7. Interpretation of Findings

The analysis supports the hypothesis that you can drive the ball as far as you want but if you can't pay it off on the greens then it may not matter.

This aligns with modern golf analytics, where strokes gained putting is a major predictor of success.

8. Limitations

Several limitations should be considered:

1. Dataset Scope
 - Limited to specific tournaments and metrics
 - May not generalize to all PGA events
2. Feature Limitations
 - Some advanced metrics may not be included
3. Model Assumptions
 - Logistic Regression assumes linear relationships
 - Tree-based models may overfit without tuning
4. External Factors Not Included
 - Weather conditions
 - Course difficulty
 - Player fatigue or injury

9. Ethical Considerations

- Data Bias: Results depend heavily on the dataset used, biased data could lead to misleading conclusions. For example, overvaluing driving metrics could lead to biased player evaluation.
- Interpretation Risk: Overemphasizing driving could undervalue other important aspects of performance
- Use of Predictions: Models should not be used as sole decision-makers in

professional contexts

10. Conclusion

This project demonstrates that:

- Machine learning models can effectively predict tournament success
- Driving metrics are more influential than putting in determining success, but putting has more of a direct impact on winning.

The workflow including preprocessing, experimentation, and evaluation ensures that the analysis is reproducible and extensible for future research.

**Github Copilot was used for debugging and Chat GPT 5.2 was also used for our report to grammar check and improve the flow and readability of the text. All analysis, interpretations, and final thoughts were completed as a group.*